Extreme Value Distribution for Global Temperatures

RIT GRADUATE SHOWCASE RESEARCH | CREATE | INNOVATE

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Problem / Question

Whether global temperature follows Extreme Value Distribution?

Has our climate been getting more extreme?

Hypothesis

- Global temperature follows Extreme Value Distribution
- The pattern is getting more extreme during more recent years

Project Overview

- The objective of this study is to fit the GEV model for global temperature data from NOAA, and try to obtain an efficient probabilistic approach from the family of extreme value distribution models, which can suitably model the uncertainties pattern in extreme global temperature event.
- The processes are:
- Science-mapping for global temperature to see the overall pattern
- Used maximum likelihood method to estimate location, scale and shape parameters and fit the GEV to the data
- Plotted time-series for the data

Data Dimensions

Longtitude

- Name: lon
- Size: 720
- Long_name: 'Longtitude
- Units: 'degree_east'
- axis: 'X'
- actual_range: array([2.5000e-01, 3.5975e+02], dtype=float32)

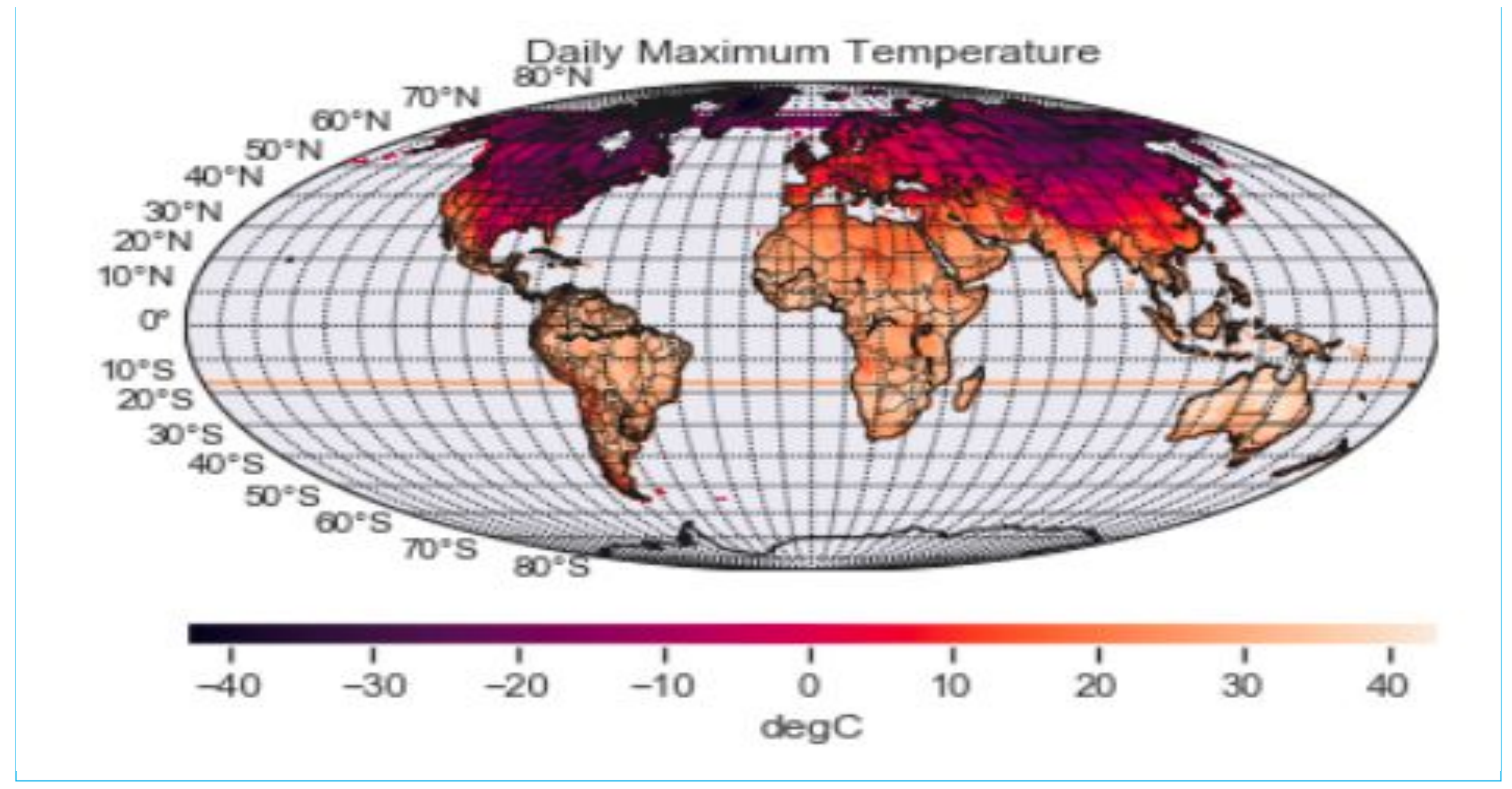
Latitude

- Name: lat
- Size: 360
- Long_name: 'Latitude
- Units: 'degree_north'
- axis: 'Y'
- actual_range: array([89.75, -89.75], dtype=float32)

Time

- Size: 365
- Long_name: 'Time'
- Units: 'hours since
- 1900-01-01 00:00:00'
- axis: 'T'
- actual_range: array([1034376.,1043112.])

Global Temperatures



Procedure

Step 1

Used Python packages to retrieve data

Step 2

Created maps for global temperatures

Step 3

Fit extreme value distribution models

Step 4

Plotted time-series and scatter plots

Data / Observations

CPC Global Daily Temperature:

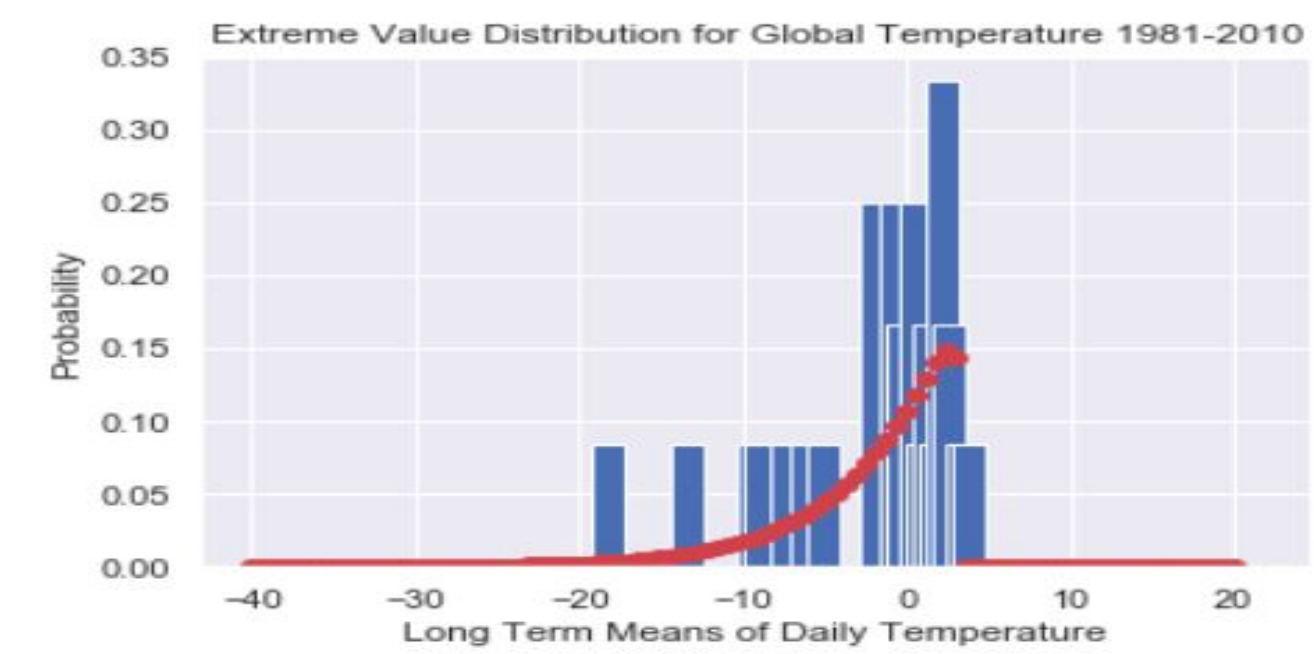
- Long Term Means of daily, monthly for years from 1981 to 2010
- Daily Maximum Temperature 2010-2019
- 2019 Daily Maximum Temperature

Works Cited

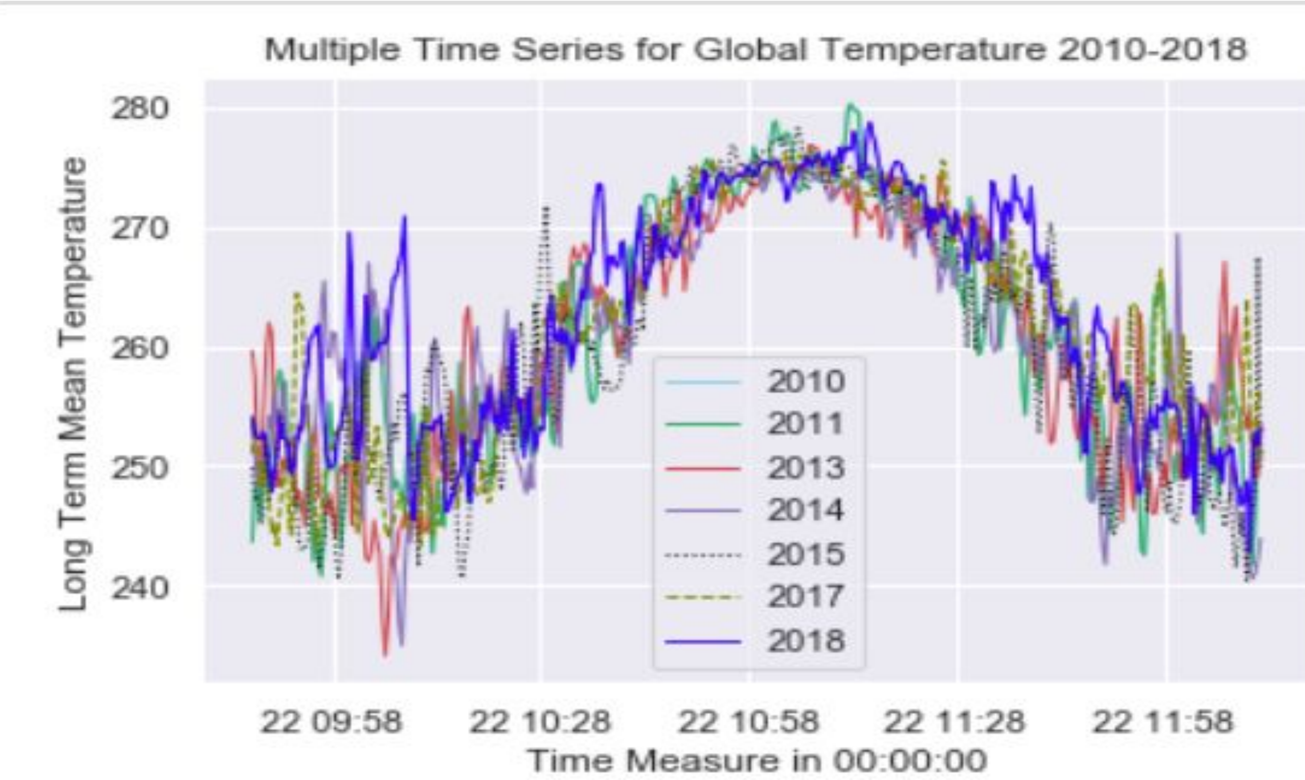
- Andre R Erler and W Richard Peltier. "Clustering of station observations for extreme value analysis". In: Proc. Fifth Int. Workshop on Climate Informatics (CI 2015) 2015 (cit. on pp. 1, 8, 11).
- W May. "Simulation of the variability and extremes of daily rainfall during the Indian summer monsoon for present and future times in a global time-slice experiment". In: Climate Dynamics 22.2-3 (2004), pp. 183–204 (cit. on pp. 1 sq.).

Results

```
Shape Parameter is 0.941830383418343
Location Parameter is -2.509716860216998
Scale Parameter is 5.291777014216436
```



- CPC Global Temperature follows Extreme Value Distribution, showing it is a highly left-skewed GEV.
- There are several outliers from the scatter plot.
- From the time-series, it is shown the pattern is getting extreme since 2015 onward comparing to previous years, and it is hotter in the summer and colder in the winter.



Conclusion

- 1981-2010 Global Temperature fits GEV, highly left-skewed. There are some outliers that might be extreme climate events.
- From time-series, it is shown that since 2015 onward, it seems hotter during summer times and colder during winter times.
- There are some missing data in CPC Global Temperature, and 2012 and 2016 has 366 days in a year, so these two years are not plotted together in the time-series.